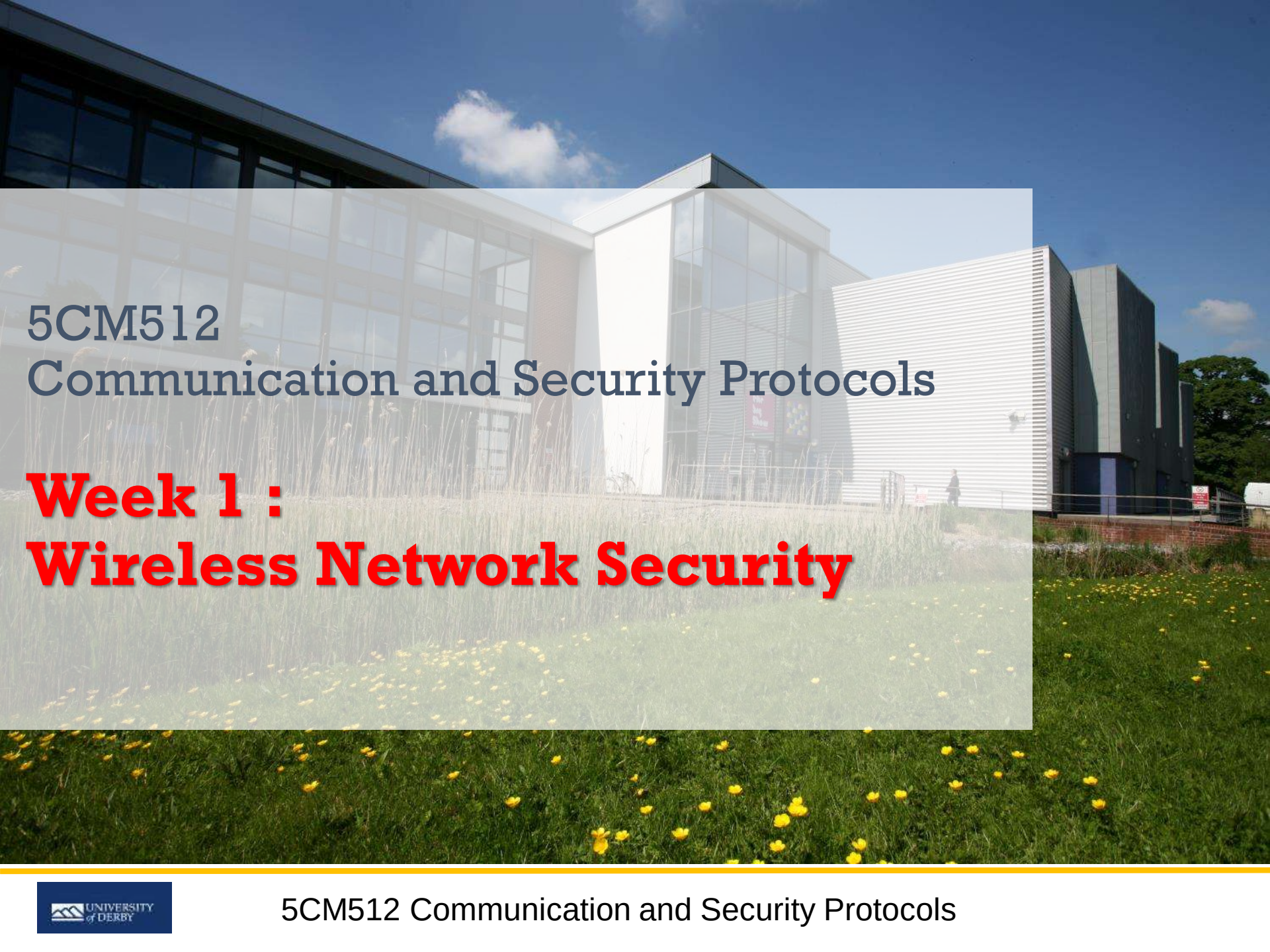


A photograph of a modern university building with large glass windows and a white facade, set against a blue sky with a few clouds. In the foreground, there is a green lawn with many yellow dandelions. A semi-transparent white box is overlaid on the left side of the image, containing the course title and week information.

5CM512

Communication and Security Protocols

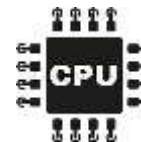
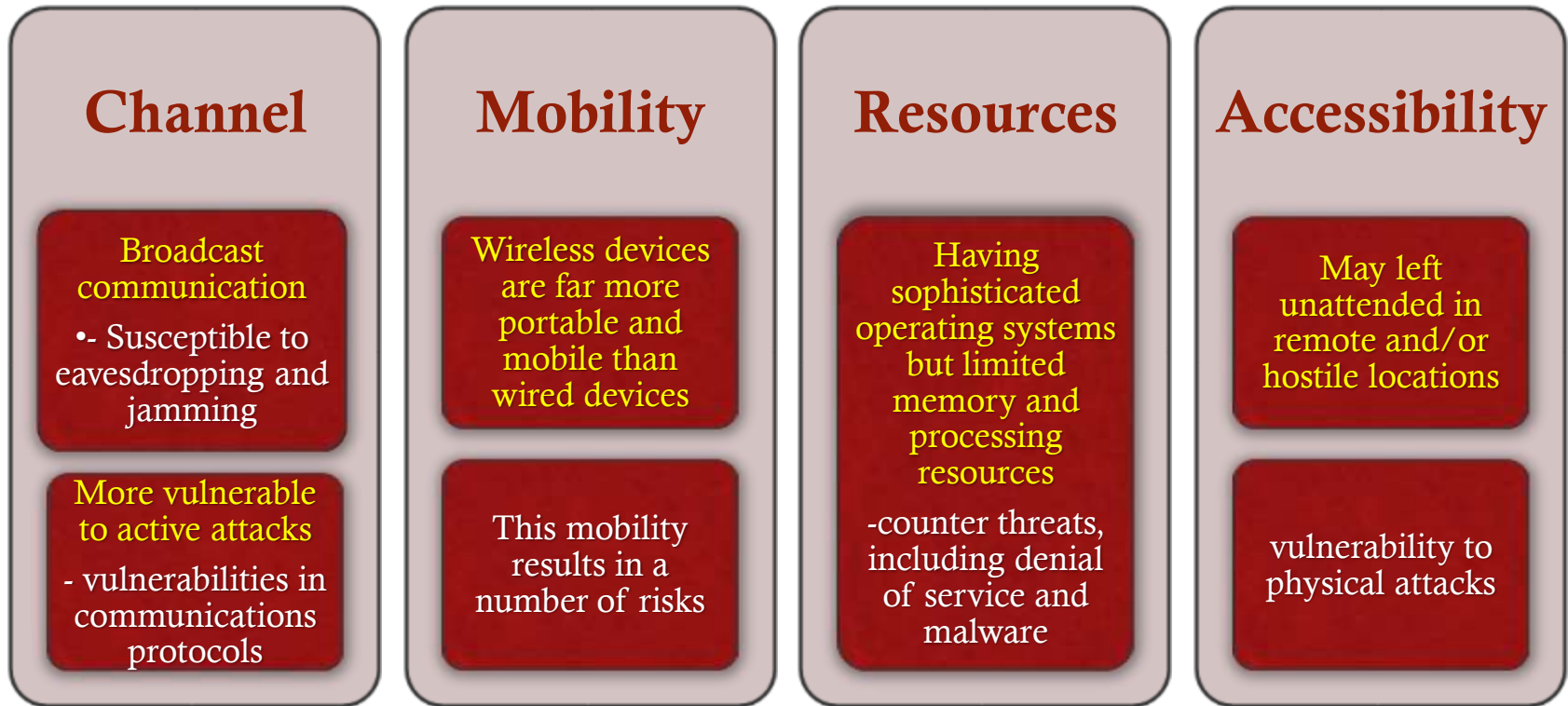
Week 1 :
Wireless Network Security

Class discussion

- Do we need special care for Wireless Networks in terms of security?
 - Can we use our security protocol developed for Wired Networks for Wireless Networks?

Wireless Security

- Key factors contributing to the higher security risk of wireless networks :



Class discussion

- What are the key components in Wireless Networking ?

Class discussion



Endpoint



Wireless medium



Access point

Wireless Network Threats

Accidental association

- Company wireless LANs in close proximity may create overlapping transmission ranges
- A user intending to connect to one LAN may unintentionally lock on to a wireless access point from a neighboring network

Malicious association

- In this situation, a wireless device is configured to appear to be a legitimate access point, enabling the operator to steal passwords from legitimate users and then penetrate a wired network through a legitimate wireless access point

Ad hoc networks

- These are peer-to-peer networks between wireless computers with no access point between them
- Such networks can pose a security threat due to a lack of a central point of control

Nontraditional networks

- Personal network Bluetooth devices, barcode readers, and handheld PDAs pose a security risk in terms of both eavesdropping and spoofing

Identity theft (MAC spoofing)

- This occurs when an attacker is able to eavesdrop on network traffic and identify the MAC address of a computer with network privileges

Man-in-the-middle attacks

- This attack involves persuading a user and an access point to believe that they are talking to each other when in fact the communication is going through an intermediate attacking device
- Wireless networks are particularly vulnerable to such attacks

Denial of service (DoS)

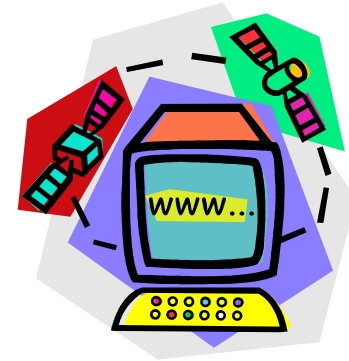
- This attack occurs when an attacker continually bombards a wireless access point or some other accessible wireless port with various protocol messages designed to consume system resources
- The wireless environment lends itself to this type of attack because it is so easy for the attacker to direct multiple wireless messages at the target

Network injection

- This attack targets wireless access points that are exposed to nonfiltered network traffic, such as routing protocol messages or network management messages

Securing Wireless Transmissions

- Principal threats to wireless transmission
 - Eavesdropping
 - Altering or inserting messages
 - Disruption
- To deal with eavesdropping, two types of countermeasures are appropriate:
 - Signal-hiding techniques
 - Turn off SSID broadcasting by wireless access points
 - Assign cryptic names to SSIDs
 - Reduce signal strength to the lowest level that still provides requisite coverage
 - Locate wireless access points in the interior of the building, away from windows and exterior walls
 - Encryption
 - Is effective against eavesdropping to the extent that the encryption keys are secured



Securing Wireless Access Points

- The main threat
 - Unauthorized access to the network
- The principal approach for preventing such access
 - IEEE 802.1x standard for port-based network access control
 - The standard provides an authentication mechanism for devices wishing to attach to a LAN or wireless network
 - The use of 802.1x can prevent rogue access points and other unauthorized devices from becoming insecure backdoors



CLASS DISCUSSION

- How can we secure wireless Networks?

Securing Wireless Networks

Use encryption

Use antivirus, antispyware software and a firewall

Turn off identifier broadcasting

Change the identifier on your router from the default

Change your router's pre-set password for administration

Allow only specific computers to access your wireless network



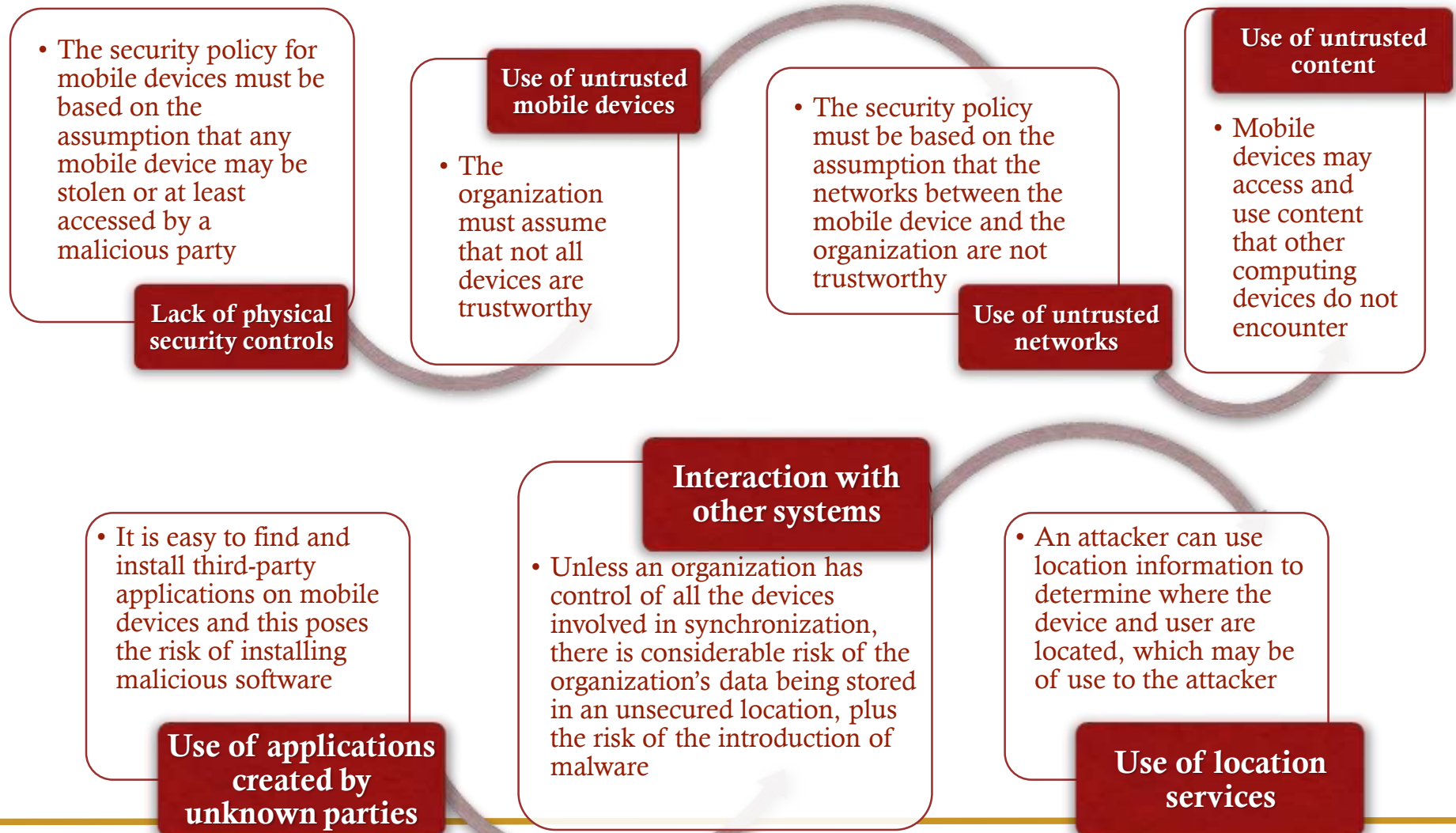
Mobile Device Security

- Mobile devices
 - Essential element for organizations as part of the overall network infrastructure
- Prior to the widespread use of smartphones, network security was based upon clearly defined perimeters that separated trusted internal networks from the untrusted Internet
- Due to massive changes, an organization's networks must now accommodate:
 - Growing use of new devices
 - Cloud-based applications
 - De-perimeterization
 - External business requirements

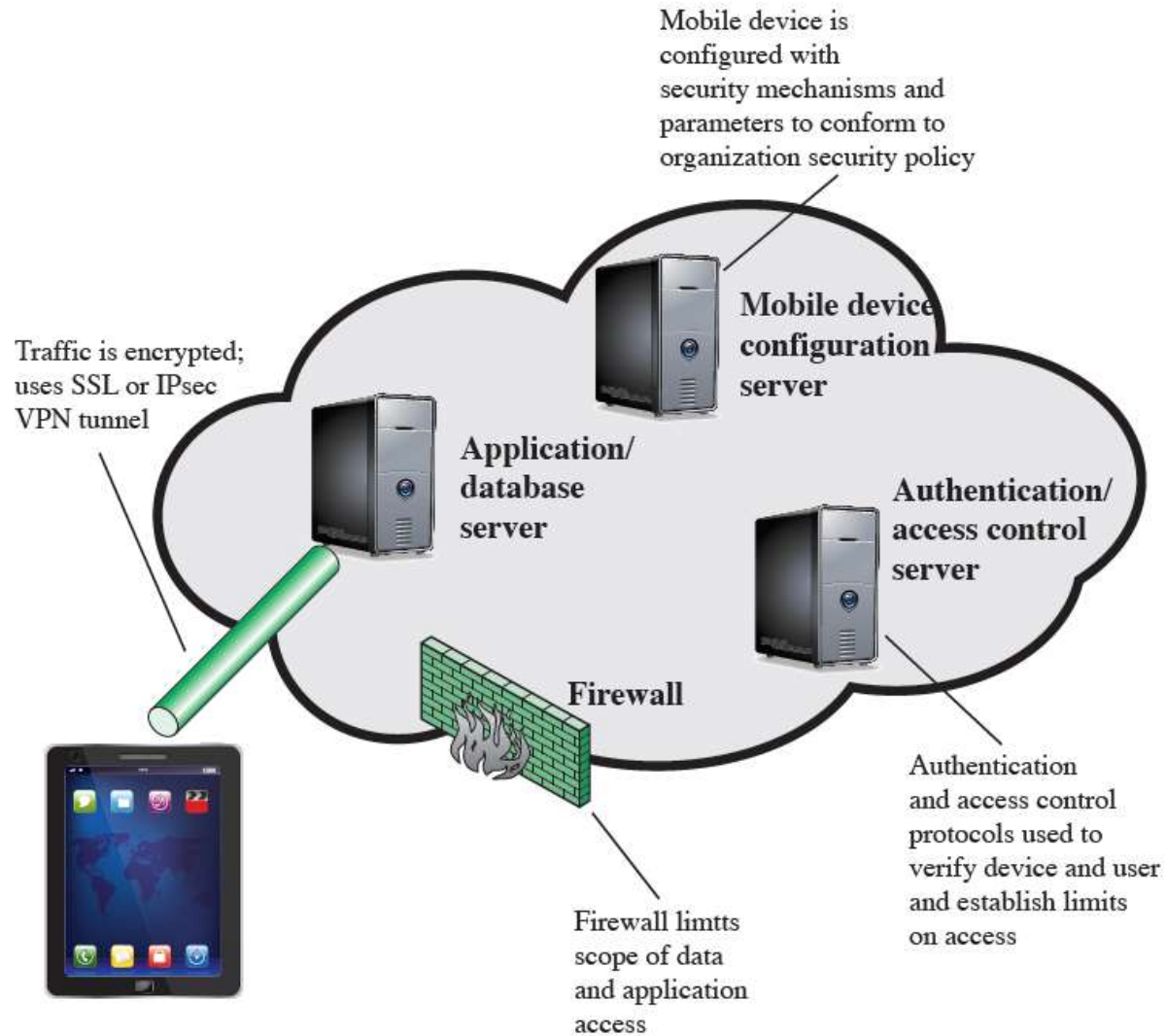


Security Threats

- Major security concerns for mobile devices:



Mobile Device Security Elements



IEEE 802.11 Wireless LAN

- Terminology

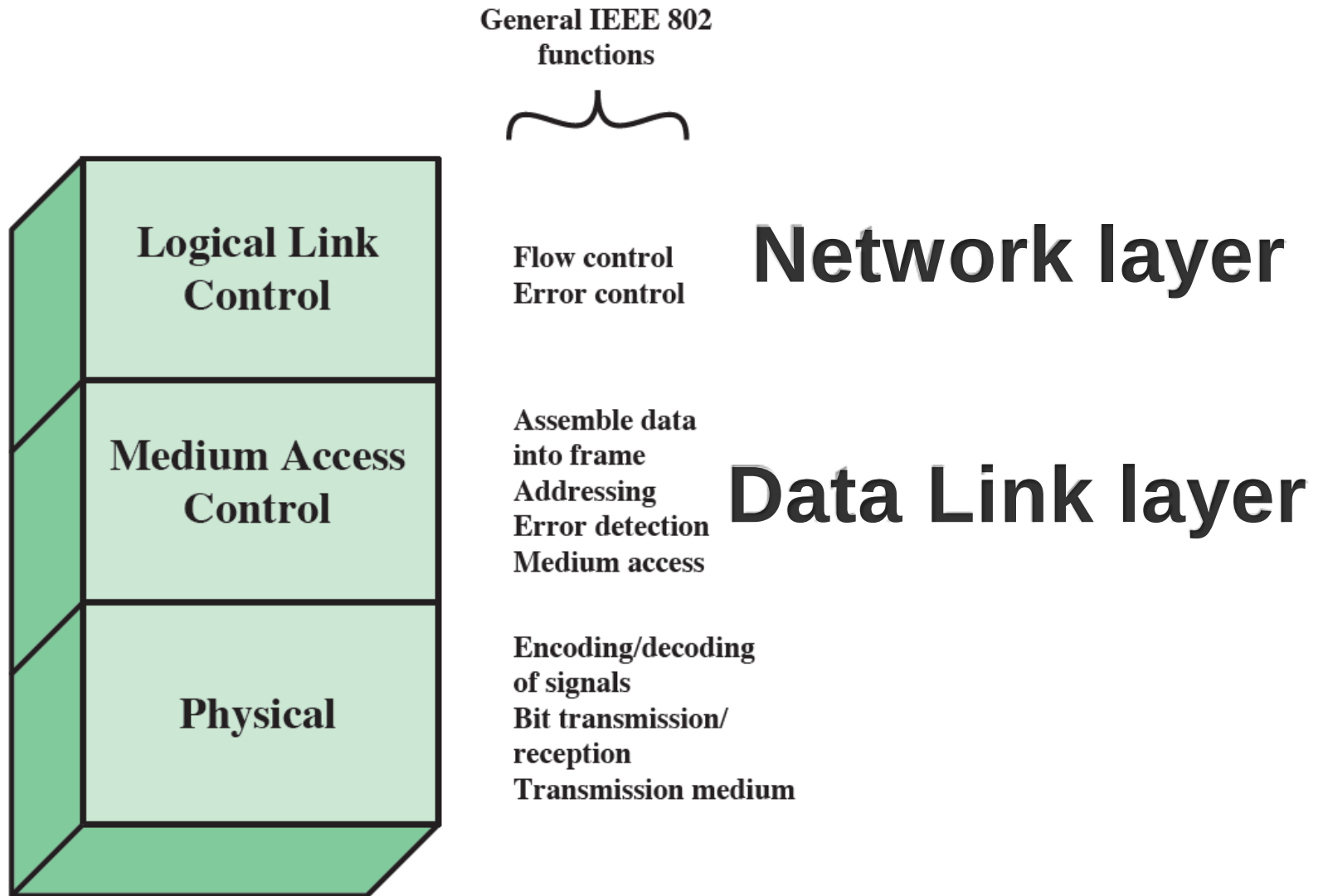
Access point (AP)	Any entity that has station functionality and provides access to the distribution system via the wireless medium for associated stations.
Basic service set (BSS)	A set of stations controlled by a single coordination function.
Coordination function	The logical function that determines when a station operating within a BSS is permitted to transmit and may be able to receive PDUs.
Distribution system (DS)	A system used to interconnect a set of BSSs and integrated LANs to create an ESS.
Extended service set (ESS)	A set of one or more interconnected BSSs and integrated LANs that appear as a single BSS to the LLC layer at any station associated with one of these BSSs.
MAC protocol data unit (MPDU)	The unit of data exchanged between two peer MAC entities using the services of the physical layer.
MAC service data unit (MSDU)	Information that is delivered as a unit between MAC users.
Station	Any device that contains an IEEE 802.11 conformant MAC and physical layer.

Wi-Fi Alliance

- The first 802.11 standard to gain broad industry acceptance was 802.11b
- Wireless Ethernet Compatibility Alliance (WECA)
 - An industry consortium formed in 1999
 - Subsequently renamed the Wi-Fi (Wireless Fidelity) Alliance
 - Created a test suite to certify interoperability for 802.11 products
- Wi-Fi
 - The term used for certified 802.11b products
 - Has been extended to 802.11g, 802.11n, 802.11ac wave-1, 802.11ac wave-2, 802.11ax (channel bandwidth 20, 40, 80, 160MHz with maximum of 2.4Gbps data rate) products
- Wi-Fi5
 - A certification process for 802.11a products that was developed by the Wi-Fi Alliance
 - The Wi-Fi Alliance has developed certification procedures for IEEE 802.11 security standards
 - Referred to as Wi-Fi Protected Access (WPA)
 - WPA



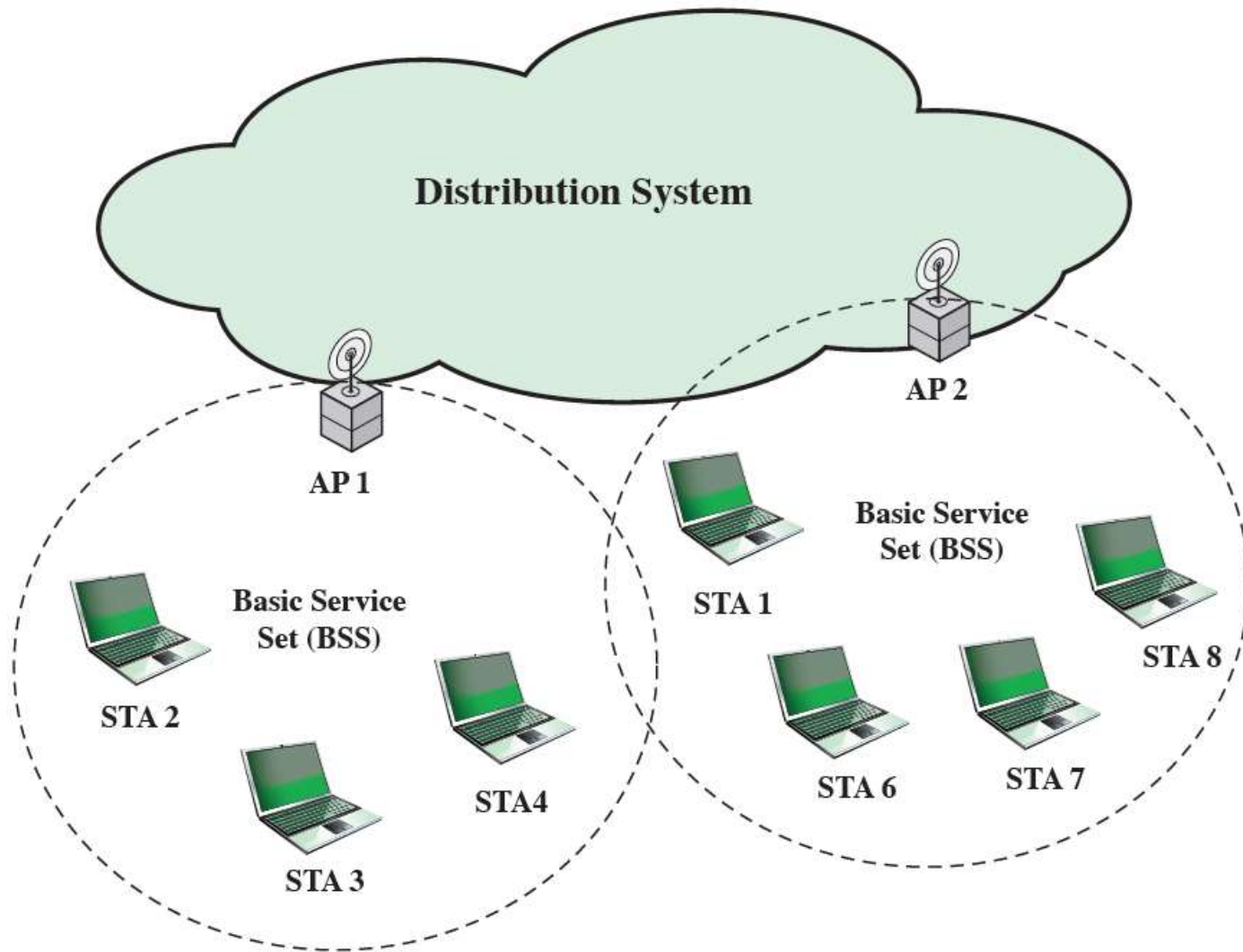
IEEE 802.11 Protocol Stack



General IEEE 802 MAC Protocol Data Unit (MPDU) Format



IEEE 802.11 Extended Service Set

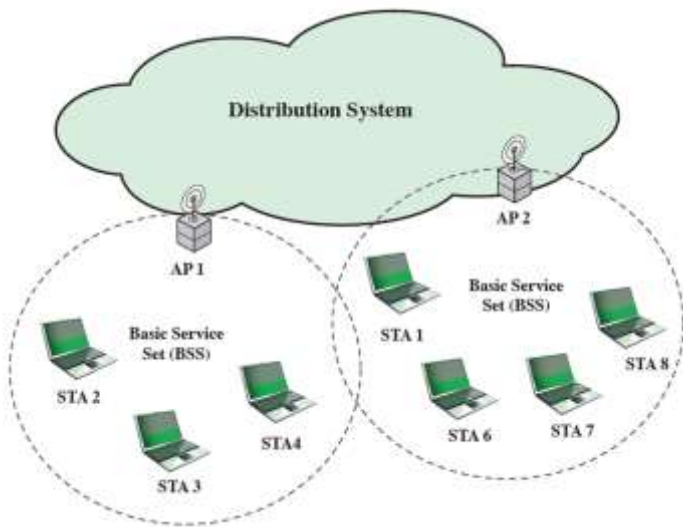
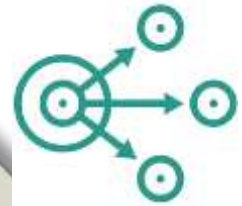


IEEE 802.11 Services

Service	Provider	Used to support
Association	Distribution system	MSDU delivery
Authentication	Station	LAN access and security
Deauthentication	Station	LAN access and security
Disassociation	Distribution system	MSDU delivery
Distribution	Distribution system	MSDU delivery
Integration	Distribution system	MSDU delivery
MSDU delivery	Station	MSDU delivery
Privacy	Station	LAN access and security
Reassociation	Distribution system	MSDU delivery

Distribution of Messages Within a DS

- The two services involved with the distribution of messages within a DS are:



Integration

- Enables transfer of data between a station on an IEEE 802.11 LAN and a station on an integrated IEEE 802.x LAN
- Takes care of any address translation and media conversion logic required for the exchange of data

Distribution

- The primary service used by stations to exchange MPDUs when the MPDUs must traverse the DS to get from a station in one BSS to a station in another BSS

Association-Related Services

- Transition types based on mobility:

No transition

- A station of this type is either stationary or moves only within the direct communication range of the communicating stations of a single BSS

BSS transition

- This is defined as a station movement from one BSS to another BSS within the same ESS
- In this case, delivery of data to the station requires that the addressing capability be able to recognize the new location of the station

ESS transition

- This is defined as a station movement from a BSS in one ESS to a BSS within another ESS
- Maintenance of upper-layer connections supported by 802.11 cannot be guaranteed
- Disruption of service is likely to occur

Association-Related Services

- To deliver a message within a DS, the distribution service needs to know the identity of the AP to which the message should be delivered in order for that message to reach the destination station
- Three services relate to a station maintaining an association with the AP within its current BSS:
 - Association
 - Establishes an initial association between a station and an AP
 - Reassociation
 - Enables an established association to be transferred from one AP to another, allowing a mobile station to move from one BSS to another
 - Disassociation
 - A notification from either a station or an AP that an existing association is terminated

Any QUESTIONS?

